

Alexandra Seceleanu

Curriculum Vitae

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Education

- 2005–2011 **PhD in Mathematics**, *University of Illinois*, Urbana–Champaign.
- 2003–2005 **Master of Science**, *Școala Normală Superioară București (SNSB)*, Romania.
Concentration in Algebra–Geometry–Topology
- 2000–2005 **Bachelor of Science**, *University of Bucharest*, Romania.
Majors: Mathematics and Computer Science

Academic Appointments

- 2026–present **Professor**, *University of Nebraska–Lincoln*, Lincoln, Nebraska.
- 2025–2026 **Milton Mohr Associate Professor**, *University of Nebraska–Lincoln*, Lincoln, Nebraska.
- 2021–2026 **Associate Professor**, *University of Nebraska–Lincoln*, Lincoln, Nebraska.
- 2015–2021 **Assistant Professor**, *University of Nebraska–Lincoln*, Lincoln, Nebraska.
- 2012–2015 **Edith T. Hitz Research Assistant Professor**, (*postdoctoral position*) *University of Nebraska–Lincoln*, Lincoln, Nebraska.
- 2011–2012 **Postdoctoral Research Fellow**, *University of Nebraska–Lincoln*, Lincoln, Nebraska.

Visiting Positions

- Aug–Dec 2025 **Visiting Scholar**, *Cornell University*, Ithaca, New York.
- Jan, May 2025 **Long Term Visitor**, *The Fields Institute*, Toronto, Ontario, Canada.
- Jan–Apr 2024 **Research Member**, *The Simons–Laufer Research Institute*, Berkeley, California.
- Aug–Dec 2012 **Postdoctoral Research Member**, *MSRI*, Berkeley, California.

Honors and Awards

- 2024 **Ruth I. Michler Memorial Prize** conferred by the Association for Women in Math.
- 2020 **College Distinguished Teaching Award** conferred by the College of Arts and Sciences.
- 2018 **Harold and Esther Edgerton Junior Faculty Award** conferred by the University of Nebraska for creative research, extraordinary teaching abilities, and academic promise.
- 2011 **Irving Reiner Memorial Award** for outstanding scholastic achievement in algebra.
- 2005–2008 **Illinois Distinguished Fellowship**, *University of Illinois at Urbana–Champaign*.

Grants

- 2024–2027 **NSF Investigator Grant** DMS–2401482.
- 2024–2029 **NSF RTG grant** DMS–2342256 with 5 co-PIs from UNL.
- 2023–2028 **UNL Grand Challenges Catalyst Grant** with many co-PIs from UNL.
- 2024–2027 **NSF Infrastructure Grant** DMS–2341670 (Polymath Jr.), co-PIs from CUNY Baruch.
- 2024–2025 **NSF Conference Grant** DMS–2433082 (WICA III).

- 2023–2024 **NSF Conference Grant** DMS–2324929 (WICA II), co-PI Haydee Lindo.
 2021–2024 **NSF Investigator Grant** DMS–2101225.
 2020–2023 **NSF Conference Grant** DMS-1953096 and **NSA Conference Grant** (BrianFest).
 2018–2019 **NSF Conference Grant** DMS-1811000 (CESMUR), co-PI's from UNL, Kansas State.
 2016–2021 **NSF Investigator Grant** DMS–1601024.

Publications and preprints

* denotes graduate student coauthor, † denotes undergraduate student coauthor

- 2026 50 *Boolean matrix rank via monomial ideals* (with J. Geraci*, A. Kunin), ArXiv:2509.09570.
 49 *Multicomplex configurations: a case study in Gorenstein liaison* (with P. Klein, J. Rajchgot), to appear in Proceedings of the WICA II-III Workshops, Springer, Cham, 2026.
 48 *General symmetric ideals* (with L. Şega), ArXiv:2506.15459.
 47 *Artinian Gorenstein algebras with binomial Macaulay dual generator* (with N. Altafi, R. Dinu, S. Masuti, R. Miro-Roig, N. Villamizar), ArXiv:2502.18149.
 46 *New families of Artinian Gorenstein algebras with the weak Lefschetz property* (with N. Altafi, R. Dinu, S. Masuti, R. Miro-Roig, N. Villamizar), to appear in Proceedings of the WICA II-III Workshops, Springer, Cham, 2026.
 45 *Constructions of Macaulay Posets and Macaulay Rings* (with P. Beall†, E. Boyali†, N. Chen†, E. Chlachidze†, T. T. Dao†, Y. O. Li†, F. Garvey†, M. Johnson†, N. Kuzmanovski, K. Ma†, R. Marasinghe†, S. Veerapaneni†), The Electronic Journal of Combinatorics 33(2) (2026), #P2.52
 44 *Products and powers of principal symmetric ideals* (with E. Dannetun†, B. Fang†, R. Formenti†, B. Gao†, J. Geraci*, R. Kogel†, Y. Li†, S. Mandal†, V. Rupasinghe†, D. Tran†, N. Walker†), J. Algebra Appl. 25 (2026), no. 2, Paper No. 2550320.
 2025 43 *Polarization and Gorenstein liaison* (with S. Faridi, P. Klein, J. Rajchgot), J. Lond. Math. Soc. (2) 112 (2025), no. 6, Paper No. e70319.
 42 *The combinatorial structure of symmetric shifted ideals* (with A. Costantini), J. Comb. Algebra 9 (2025), no. 3–4, 209–263.
 41 *Polynomial growth of Betti sequences over local rings* (with L. Avramov and Z. Yang*), Collect. Math. 76 (2025), no. 3, 627–650.
 40 *Higher Lorentzian polynomials, higher Hessians, and the Hodge–Riemann relations for graded oriented Artinian Gorenstein algebras in codimension two* (with P. Macias-Marques, C. McDaniel), Int. Math. Res. Not. IMRN, 2025 no. 13, 1–34.
 39 *Lefschetz properties through a topological lens*, Ann. Univ. Paedagog. Crac. Stud. Math. , 24 (2025), 7–47.
 38 *The minimal free resolution of a general principal symmetric ideal* (with M. Harada, L. Şega), Trans. Amer. Math. Soc. 378 (2025), no. 3, 1831–1882.
 37 *Betti numbers for connected sums of graded artinian Gorenstein algebras* (with N. Altafi, R. DiGennaro, F. Galetto, S. Grate*, R. Miró-Roig, U. Nagel, J. Watanabe), Trans. Amer. Math. Soc. 378 (2025), no. 2, 1055–1080.
 2023 36 *Duality for asymptotic invariants of graded families* (with M. DiPasquale, T. Nguyễn), Adv. Math. 430 (2023), Paper No. 109208.

- 35 *Lefschetz properties of some codimension three Artinian Gorenstein algebras* (with N. Abdallah, N. Altafi, A. Iarrobino, J. Yaméogo), *J. Algebra* 625 (2023), 28–45.
- 34 *Cohomological blow ups of graded artinian Gorenstein algebras along surjective maps* (with A. Iarrobino, P. Macias Marques, C. McDaniel and J. Watanabe), *Int. Math. Res. Not. IMRN* 2023, no. 7, 5816–5886.
- 33 *Axial constants and sectional regularity of homogeneous ideals* (with M. DeBellevue*, A. Lebovitz†, Y. Li†, M. Lotfi†, S. Mohite†, X. Pan†, M. S. Pathak†, S. Roshan Zamir*, S. X. Zhang†), *Proc. Amer. Math. Soc.* 151 (2023), no. 4, 1363–1378.
- 32 *Computing rational powers of monomial ideals* (with P. Dongre†, B. Drabkin, J. Lim†, E. Partida†, E. Roy†, D. Ruff†, T. Tang), *J. Symbolic Comput.* 116 (2023), 39–57.
- 2022 31 *Convex bodies and asymptotic invariants for powers of monomial ideals* (with J. Camarneiro†, B. Drabkin, D. Fragoso†, W. Frendreiss†, D. Hoffman†, T. Tang, S. Yang†), *J. Pure Appl. Algebra* 226 (2022), no. 10, paper no. 107089.
- 30 *Rees algebras of filtrations of covering polyhedra and integral closure of powers of monomial ideals* (with G. Grisalde* and R. Villarreal), *Res. Math. Sci.* 9 (2022), no. 1, paper no. 13. 13 (05).
- 29 *Connected sums of graded artinian Gorenstein algebras and Lefschetz properties* (with C. McDaniel, A. Iarrobino), *J. Pure Appl. Algebra* 226 (2022), no. 1, 106787.
- 2021 28 *Canonical resolutions over Koszul algebras* (with E. Faber, M. Juhnke-Kubitzke, H. Lindo, C. Miller, R. R.G.), *Women in Commutative Algebra—Proceedings of the 2019 WICA Workshop*, Springer, Cham, 2021.
- 27 *Symbolic Rees algebras* (with E. Grifo), *Commutative Algebra* 343–371, Springer, Cham, 2021.
- 26 *Consequences of the packing problem* (with H. Bodas†, B. Drabkin, C. Fong†, S. Jin†, J. Kim†, W. Li†, T. Tang, B. Williams†), *J. Algebraic Combin.* 54 (2021), no. 4, 1095–1117.
- 25 *Singular loci of reflection arrangements and the containment problem* (with B. Drabkin*), *Math. Z.* 299 (2021), no. 1–2, 867–895.
- 2020 24 *Quadratic Gorenstein rings with many surprising properties* (with J. McCullough), *Arch. Math. (Basel)* 115 (2020), no. 5, 509–521 – named distinguished article by the editors.
- 23 *Symbolic powers of codimension two Cohen-Macaulay ideals* (with S. Cooper, G. Fatabbi, E. Guardo, A. Lorenzini, J. Migliore, U. Nagel, J. Szpond, A. Van Tuyl), *Comm. Algebra* 48 (2020), no. 11, 4663–4680 – awarded Earl Taft Memorial Award.
- 22 *Implicitization of tensor product surfaces via virtual projective resolutions* (with E. Duarte), *Math. Comput.* (2020), 89 (2020), no. 326, 3023–3056.
- 21 *Betti numbers of symmetric shifted ideals* (with J. Biermann, H. De Alba Casillas, F. Galetto, S. Murai, U. Nagel, T. Römer, A. O’Keefe), *J. Algebra* 560 (2020), 312–342.
- 20 *Frieze varieties: A characterization of the finite-tame-wild trichotomy for acyclic quivers* (with K. Lee, L. Li, M. Mills and R. Schiffler), *Adv. Math.* 367 (2020), 107130.
- 19 *Generalized minimum distance functions and algebraic invariants of Geramita ideals* (with S. Cooper, Ş. Tohăneanu, M. Vaz Pinto and R. Villarreal), *Adv. in Appl. Math.* 112 (2020) 101940.

- 2019 18 *Negative curves on symmetric blowups of the projective plane, resurgences and Waldschmidt constants* (with Th. Bauer, S. Di Rocco, B. Harbourne, J. Huizenga, T. Szemberg), *Int. Math. Res. Not. IMRN* 2019, no. 24, 7459–7514.
- 17 *Calculations involving symbolic powers* (with B. Drabkin*, E. Grifo*, B. Stone), *JSAG Journal of Software for Algebra and Geometry*, vol. 9, no.1 (2019), 71–80.
- 16 *Determinants of incidence and Hessian matrices arising from the vector space lattice* (with S. Nasseh and J. Watanabe), *J. Commutative Algebra*, vol. 11, no. 1 (2019), 131–154.
- 2018 15 *A tight bound on the projective dimension of four quadrics* (with C. Huneke, J. McCullough and P. Mantero), *J. Pure Appl. Algebra* 222 (2018), no. 9, 2524–2551.
- 2016 14 *The Waldschmidt constant for squarefree monomial ideals* (with C. Bocci, S. Cooper, E. Guardo, B. Harbourne, M. Janssen, U Nagel, A. Van Tuyl, T. Vu), *J. Algebraic Combin.* 44 (2016), no. 4, 875–904.
- 13 *Ordinary and symbolic Rees algebras of Fermat point configurations* (with U. Nagel), *J. Algebra* 468 (2016), 80–102.
- 12 *Multiple structures with arbitrarily large projective dimension on linear subspaces* (with C. Huneke, J. McCullough and P. Mantero), *J. Algebra* 447 (2016), 183–205.
- 2015 11 *Resurgences for ideals of special point configurations in $\mathbb{P}^N$ coming from hyperplane arrangements* (with M. Dumnicki, B. Harbourne, U. Nagel, T. Szemberg, H. Tutaj-Gasinska), *J. Algebra* 443 (2015), 383–394.
- 10 *A homological criterion for the failure of containment between symbolic and ordinary powers of ideals of points in $\mathbb{P}^2$* , *J. Pure Appl. Algebra* 219 (2015), no. 11, 4857–4871.
- 9 *A multiplicity bound for graded rings and a criterion for the Cohen-Macaulay property* (with C. Huneke, J. McCullough and P. Mantero), *Proc. Amer. Math. Soc.* 143 (2015), no. 6, 2365–2377.
- 8 *Containment counterexamples for ideals of various configurations of points in $\mathbb{P}^N$* (with B. Harbourne), *J. Pure Appl. Algebra* 219 (2015), 1062–1072.
- 2014 7 *Syzygies and singularities of tensor product surfaces of bidegree $(2, 1)$* (with H. Schenck and J. Validashti), *Math. Comp.* 83 (2014), 1337–1372.
- 2013 6 *The projective dimension of codimension two algebras presented by quadrics* (with C. Huneke, J. McCullough and P. Mantero), *J. Algebra* 393 (2013), 170–186.
- 5 *Bounding projective dimension* (with J. McCullough) - *Commutative Algebra*, Springer-Verlag London Ltd., London, 2013.
- 2012 4 *Syzygy theorems via comparison of order ideals on a hypersurface* (with P.A. Griffith) *J. Pure Appl. Algebra* 216 (2012), 468–479.
- 2011 3 *Inverse systems, Gelfand-Tsetlin patterns and the weak Lefschetz property* (with B. Harbourne and H. Schenck) *J. London Math. Soc.*, 84 (2011), 712–730.
- 2 *Ideals with large projective dimension and regularity* (with J. Beder, J. McCullough, L. Nunez-Betancourt, B. Snapp, B. Stone), *J. Symb. Comput.* 46 (2011), 1105–1113.
- 2010 1 *The Weak Lefschetz Property and powers of linear forms in $K[x, y, z]$* (with H. Schenck), *Proc. Amer. Math. Soc.* 138 (2010), 2335–2339.

Writing contributions to the profession

- 1 *When life gives you lemons, make mathematicians* (with K. Adaricheva, B. Brubaker, P. Devlin, S. J. Miller, V. Reiner, A. Sheffer, Y. Zeytuncu), Notices of the American Mathematical Society, March 2021.

Teaching Experience

All course listed below were taught as *instructor of record* with full course responsibilities.

At the University of Nebraska-Lincoln:

MATH 106 Calculus I (large lectures)	Fall 2023, Fall 2024
MATH 106 Calculus II (large lecture)	Spring 2026
MATH 189H The Joy of Numbers (honors course)	Fall 2014
MATH 208 Calculus III	Spring 2013
MATH 310 Introduction to Modern Algebra	8 times
MATH 314/814 Linear Algebra	3 times
MATH 314H Honors Applied Linear Algebra (honors course)	Spring 2017
MATH 325 Introduction to Real Analysis	Fall 2017, Spring 2021
MATH 407 Mathematics for High School Teaching I	Spring 2018
MATH 415/815 Linear Transformations	Spring 2023
MATH 417 Group Theory	Spring 2013
MATH 435 Math in the City ²	Fall 2013
MATH 805T Discrete Mathematics for Teachers	Summer 2016
MATH 810T Algebra for Algebra Teachers	Fall 2026
MATH 811T Functions for High School Teachers	Fall 2025
MATH 817 Introduction to Modern Algebra I (graduate course)	Fall 2018, 2022
MATH 818 Introduction to Modern Algebra II (graduate course)	Spring 2017, 2019
MATH 901 Modern Algebra I (graduate course)	Fall 2020
MATH 902 Modern Algebra II (graduate course)	Spring 2020, 2021
MATH 905 Commutative Algebra (graduate course)	Fall 2017, 2026
MATH 918 Computational Algebra (graduate topics course)	Fall 2013
MATH 918 Geometry of Syzygies (graduate topics course)	Spring 2016
MATH 918 The Lefschetz Properties (graduate topics course)	Fall 2019
MATH 953 Algebraic Geometry (graduate course)	Spring 2025

At the University of Illinois at Urbana-Champaign:

MATH 231 Calculus II, <i>Calculus & Mathematica</i> ¹	Fall 2008, Fall 2009
MATH 181 A Mathematical World	Summer 2006

¹ Course uses *Mathematica* to aid visualization and develop conceptual understanding.

² Capstone course on topics in mathematical modeling in collaboration with local institutions.

Teaching Development and Awards

- 2020 College Distinguished Teaching Award, UNL College of Arts and Sciences.
- 2018 Roger Wiegand Award for contributions to the UNL mathematics graduate program.
- 2017–2018 Completed the *Peer Review of Teaching Project*, a year-long program centered around (re)designing a course, implementing it and constructing a teaching portfolio.
- 2014 Certificate of Recognition for Contributions to Students recipient, UNL.

2008,2009 Appeared on the List of Teachers Ranked as Excellent by their students, UIUC.

Undergraduate Research Supervised

- 2025 *Investigating quantum circuits using polynomials* – Bishop Placke, supported by UCARE
- 2025 *International REU in Commutative Algebra: Binomial Edge Ideals* – 20 students
- 2024 *Polymath Jr: Macaulay Rings and Macaulay Posets* – 25 students
- 2023–2024 *Abstract Algebra and U.S. National Security: Transitioning Use of the RSA Algorithm in Post-Quantum Cryptography*
 - honors thesis by Helena Holland awarded highest distinction
- 2023 *Polymath Jr: Strength and Symmetry of Polynomials* – 23 students
- 2021 *Hessian matrices in three dimensions* – Turner Blick, supported by a UCARE grant
- 2021 *Polymath Jr: Computational Algebra* – 22 students
- 2020 *Polymath Jr: Monomials, Convex Bodies and Optimization* – 29 students
- 2017 *The Waldschmidt constant and symbolic powers of monomial ideals* – Diana Zhong
- 2015 *Tropical rank for phylogenetic metrics* – Cashous Bortner, Kaylee Homolka, William Morgan, Jack Rodenburg
- 2013–2015 *Unimodality for independence sequences of well-covered graphs of small type*
 - honors thesis by Joey Becker – supported by a UCARE grant
- 2014 *Sustainability of Nebraska Groundwater Reservoirs* – Angie Loong, Katie Pawlowski, Helen Pitts, Jacqui Voss and Shalima Zalsha

Graduate Students Supervised

- 2016–present Thesis advisor and supervisory committee (co-)chair for
 - Benjamin Drabkin (Ph.D. 2020)
 - Andrew Conner (M.S. 2020)
 - Erica Hopkins (Ph.D. 2021), co-advised with Mark Walker
 - Michael DeBellevue (Ph.D. 2022), co-advised with Mark Walker
 - Nikola Kuzmanovski (Ph.D. 2024), co-advised with Jamie Radcliffe
 - Shah Roshan Zamir (Ph.D. 2025)
 - Juliann Geraci (since 05/2022), co-advised with Alexander Kunin (Creighton)
 - Ana Podariu (since 04/2024)
 - Kara Fagerstrom (since 07/2024)
 - Ben Huenemann (since 04/2025)
 - Noah Walker (since 12/2025)

Postdoctoral Mentoring

- 2021–2022 Alessandra Costantini, supported by an AWM mentoring travel grant.
- 2023–present Lauren Cranton Heller, supported by an NSF postdoctoral grant fall 2025–present.

Selected Presentations

Lecture Series

- 05/2024 Summer school on Lefschetz Properties, Pedagogical University, Krakow, (3 lectures).
- 01/2024 Introductory Workshop to the Commutative Algebra program, SLMath, (3 lectures).

- 06/2023 National School on Algebra, University of Bucharest, (4 lectures).
[Invited Presentations](#)
- 07/2026 MEGA – Effective Methods in Algebraic Geometry (plenary speaker), Durham, UK.
 06/2026 Commutative Algebra and Algebraic Geometry in Milan, Italy.
 10/2025 AMS Sectional Meeting, Tulane University.
 08/2025 Cornell University mathematics colloquium.
 06/2025 Jordan Types of Artinian Algebras and Geometry of Punctual Hilbert Schemes, Université Côte d'Azur, Nice, France.
 05/2025 AMS Sectional Meeting, California Polytechnic University, San Luis Obispo.
 02/2025 TATERS seminar organized by Boise State University, (online).
 01/2025 Commutative Algebra meets Algebraic Combinatorics, Fields Institute / U. of Toronto.
 10/2024 Open Problems seminar, Stockholm University, (online).
 08/2024 Commutative Algebra and its Connections to Algebraic Geometry and Combinatorics, Notre Dame University.
 12/2023 Syracuse University mathematics colloquium.
 10/2023 AMS Sectional Meeting, Creighton University, Omaha.
 10/2023 Fellowship of the Ring international commutative algebra seminar.
 09/2023 AWM Symposium, Clark Atlanta University, Atlanta.
 05/2023 MSRI graduate school colloquium, Notre Dame University.
 05/2023 Lefschetz Properties in Algebra, Geometry, Topology and Combinatorics, Fields Institute, Toronto, Canada.
 03/2023 IIT Bombay international commutative algebra seminar, (online).
 03/2023 Cleveland State University mathematics colloquium.
 01/2023 AMS Joint Mathematics Meeting, Boston.
 01/2023 CIRM – Commutative Algebra – Written Geometry, Luminy, France.
 11/2022 MFO – General Projections of Algebraic Varieties, Oberwolfach, Germany.
 09/2022 INdAM – The Strong and Weak Lefschetz Properties, Cortona, Italy.
 04/2022 CIMAT Commutative Algebra & Algebraic Geometry Seminar, Guanajuato, Mexico.
 03/2022 AMS Sectional Meeting, (online).
 02/2022 Iranian Commutative Algebra Seminar, (national online seminar), Tehran, Iran.
 02/2022 MFO – Interpolation, Approximation and Algebra, Oberwolfach, Germany.
 02/2022 Women in Commutative Algebra and Algebraic Geometry, (online), Fields Institute.
 11/2021 AMS Sectional Meeting, (online).
 08/2021 SIAM Conference on Applied Algebraic Geometry, (online).
 05/2021 AMS Sectional Meeting, (online).
 01/2021 AMS-MAA Joint Mathematics Meeting.
 12/2020 Cleveland State University mathematics colloquium, (online).
 12/2020 Algebra, Geometry, and Combinatorics colloquium, (national online seminar).
 09/2020 AMS Sectional Meeting.
 07/2020 Fellowship of the ring national commutative algebra seminar, organized by MSRI.

10/2019 CIRM – Lefschetz properties in Algebra, geometry and Combinatorics II, Luminy, France.
 07/2019 SIAM Conference on Applied Algebraic Geometry, Bern, Switzerland.
 04/2019 Graduate Workshop in Commutative Algebra for Women and Mathematicians of Other Minority Genders, Minneapolis, MN.
 04/2019 AWM Research Symposium, Houston, TX.
 12/2018 Canadian Math Society winter meeting, Vancouver, BC, Canada.
 11/2018 Geometric Methods in Representation Theory conference, Iowa City, IA.
 10/2018 MFO – Asymptotic Invariants of Homogeneous Ideals, Oberwolfach, Germany.
 06/2018 BIRS workshop – New Trends in Syzygies, Banff, Canada.
 04/2018 Commutative Algebra workshop, Charlottesville, VA.
 03/2018 AMS Sectional Meeting, Columbus, OH.
 10/2017 Structures on Free Resolutions, Lubbock, TX.
 09/2017 AMS Sectional Meeting, Denton, TX.
 08/2017 SIAM Conference on Applied Algebraic Geometry, Atlanta, GA.
 07/2017 Stillman’s Conjecture and Other Progress on Free Resolutions, Berkeley, CA.
 06/2017 Commutative Algebra Meeting Algebraic Geometry, Bucharest, Romania.
 05/2017 CMO – Symbolic and Ordinary Powers of Ideals, Oaxaca, Mexico.
 08/2016 CMO – Geometric Modeling and Approximation Theory, Oaxaca, Mexico.
 07/2016 Commutative Algebra and its Interactions with Algebraic Geometry: Tight Closure, Linkage, and Syzygies, Ann Arbor, MI.
 04/2016 AMS Sectional Meeting, Fargo, ND.
 04/2016 AMS Sectional Meeting, Salt Lake City, UT.
 03/2016 AMS Sectional Meeting, Athens, GA.
 02/2016 MFO – Arrangements of subvarieties & their applications in algebraic geometry, Oberwolfach, Germany.
 10/2015 KUMUNU, University of Missouri, Columbia, MO.
 07/2015 Eighth International Congress of Romanian Mathematicians, Iași, Romania.
 07/2015 Minimal free resolutions, Betti numbers, and combinatorics, Edinburgh, Scotland.
 02/2015 MFO–Linear systems, symbolic powers and Waring problems, Oberwolfach, Germany.
 01/2015 AMS-MAA Joint Mathematics Meeting, San Antonio, TX.
 09/2014 AMS Sectional Meeting, Eau-Claire, WI.
 01/2014 AMS-MAA Joint Mathematics Meeting, Baltimore, MD.
 10/2013 AMS Sectional Meeting, Louisville, KY.
 10/2013 AMS Sectional Meeting, Louisville, KY.
 09/2013 Interactions between Commutative Algebra and Algebraic Geometry II.
 Tulane University, New Orleans, LA
 08/2013 SIAM Conference on Applied Algebraic Geometry, Fort Collins, CO.
 06/2013 AMS - RMS Joint International Meeting, Alba-Iulia, Romania.
 11/2012 Workshop on Syzygies, Cornell University, Ithaca, NY.
 09/2012 Focus Period Members’ Seminar, *MSRI*, Berkeley.

- 04/2012 Algebra, Geometry and Combinatorics Day, *UIUC*, Champaign, IL.
- 03/2012 AMS Sectional Meeting, Lawrence, KS.
- 10/2011 AMS Sectional Meeting, Lincoln, NE.
- 07/2011 Seventh International Congress of Romanian Mathematicians, Braşov, Romania.
- 05/2011 Resolutions Day, Cornell University, Ithaca, NY.
- 01/2011 AMS-MAA Joint Mathematics Meeting, New Orleans, LA .
- 01/2010 AMS-MAA Joint Mathematics Meeting, San Francisco, CA.
- 11/2009 AMS Sectional Meeting, Boca Raton, FL.

Conference Organization

- 2025 Special session on *Algebraic and Homological Perspectives with a view towards Geometry*, AMS Sectional Meeting, University of Kansas.
- 2024 *Women in Commutative Algebra III* workshop at Casa Matemática Oaxaca, Mexico.
- 2024 *Connections* workshop of the Commutative Algebra thematic program, MSRI, Berkeley.
- 2024 Special session on *Group Actions in Commutative Algebra*, Joint Mathematics Meetings, San Francisco.
- 2024 Special session on *Polymath Jr REU Student Research*, Joint Mathematics Meetings, San Francisco.
- 2023 *Women in Commutative Algebra II* workshop, CIRM, Trento, Italy.
- 2023 Research conference on *Unexpected and Asymptotic Properties of Projective Varieties*, University of Nebraska–Lincoln.
- 2021 Special session on *Commutative Algebra*, AMS Sectional Meeting, Creighton University.
- 2019 Special session on *Recent Advances in Homological and Commutative Algebra*, Joint Mathematics Meetings, Baltimore, MD.
- 2018 Focused research group on *Investigating Linear Codes via Commutative Algebra*, BIRS, Banff, Canada.
- 2017 Special session on *Commutative Algebra*, AWM Research Symposium, UCLA.
- 2014 Special session on *Commutative Algebra and Algebraic Geometry*, AMS Sectional Meeting, Texas Tech University.

Service to the Profession

- since 2025 Editor for the diamond open access journal Combinatorial Commutative Algebra.
- 2025 Editor for the Proceedings of the WICA II-III Conferences, to be published by Springer.
- since 2024 Editor for the Journal of Commutative Algebra.
- since 2017 Reviewer for the NSF Algebra and Number Theory Grant Program.
- Reviewer for the NSF Graduate Research Fellowships Program.
- Reviewer for the NSA Mathematical Sciences Grant Program.
- Reviewer for the Simons Foundation Collaboration Grants for Mathematicians Program.
- Reviewer for the Banff International Research Station, Canada.
- Reviewer for the Alexander von Humboldt Foundation fellowship program.
- Reviewer for the National Science Center, Poland.

Reviewer for the Louisiana Board of Regents.

since 2011 Referee for the following journals :

- Algebra and Number Theory
- Archiv der Mathematik
- Advances in Applied Mathematics
- Bulletin of the London Mathematical Society
- Canadian Journal of Mathematics
- Compositio Mathematica
- Contemporary Mathematics
- Collectanea Mathematica
- European Journal of Combinatorics
- Geometriae Dedicata
- Illinois Journal of Mathematics
- International Journal of Algebra and Computation
- International Mathematics Research Notices
- Israel Journal of Mathematics
- Journal of Algebra
- Journal of Algebra and Its Applications
- Journal of Algebraic Combinatorics
- Journal of Commutative Algebra
- Journal of Combinatorial Theory Series A
- Journal of the London Mathematical Society
- Journal of Pure and Applied Algebra
- Journal of Software for Algebraic Geometry
- Journal of Symbolic Computation
- Mathematics of Computation
- Mathematische Nachrichten
- Mathematische Zeitschrift
- MethodsX
- Michigan Mathematical Journal
- Portugaliae Mathematica
- Proceedings of the American Mathematical Society
- Results in Mathematics
- SIAM Journal on Applied Algebra and Geometry
- Transactions of the American Mathematical Society.

University Service

- 2024-25 Mathematics Department Awards Nomination Committee.
- 2017-19 & Mathematics Department Executive Committee member (elected).
- 2023–2025 - *Performed faculty annual evaluations and strategic planning for the department.*
- 2012–present Undergraduate Activities Committee member (co-chair 2016–17, chair 2017–20, 2023).
 - 2023 Mathematics Department 125-year Celebration Committee member.
- 2019, 2022 Mathematics Department Tenure-track Search Committee member.
- 2022–2023 Undergraduate Program Committee
 - 2021 Focus group on envisioning the future of NSF Epscor (university-wide).
 - 2021 Edgerton award selection committee (university committee).
- 2020–2021 Graduate recruitment coordinator and Graduate Advisory Committee member.
- 2016–2022 AMS graduate student chapter faculty advisor.
- 2016–2024 Teaching mentor for several graduate students and junior faculty in mathematics.
- 2015–2017 Mathematics Department Graduate Advisory Committee member.
 - 2017 Ad-hoc Search Committee for the Chair of the Mathematics Department member.
- 2014-2017 Mentor for two undergraduate students in the Nebraska Math program
 - 2013 Undergraduate Creative Activities and Research interdisciplinary selection committee
 - 2012 Nebraska Conference for Undergraduate Women in Math—*organized faculty program.*
- 2011, 2013 Putnam contest training – *led problem solving sessions for UNL students.*

Outreach Efforts

Early–Career Researcher Mentoring

- 01/2025 Research Mentor: Apprenticeship Program, Fields Institute, Toronto, Canada

Graduate Summer Schools

- 06/2024 Lecturer: Lefschetz Properties School, Krakow, Poland
06/2023 Lecturer: National Algebra School, Bucharest, Romania

Undergraduate Research

- 2025 Co-organizer of international REU in Commutative Algebra at CIMAT, Mexico
since 2024 Founder and director of the Nebraska Experimental Math and Outreach (NEMO) Lab
- *Organized experiential research opportunities for small groups of undergraduates.*
since 2020 Management team member for the Polymath Jr. undergraduate research program,
serving ~ 300 students each year.
2017–2019 Organizer for the Central States Mathematics Undergraduate Research Conference.
- *A conference showcasing research performed by undergraduate students.*
2015–2020 Organizer for the lecture series Career Perspectives in Mathematics.
- *Featuring professional success stories from UNL mathematics alumni.*

Women in Mathematics

- 2023–2024 Organizer for the Women in Commutative Algebra WICA II and WICA III workshops.
10/2019 Project co-leader at the Women in Commutative Algebra (WICA I) workshop.
04/2019 Invited speaker at the Graduate Workshop in Commutative Algebra for Women & Mathematicians of Other Minority Genders.
02/2019 Invited speaker by the Association for Women in Mathematics student chapter at the University of Kentucky.
09/2016 Invited speaker by the Association for Women in Mathematics student chapter at the University of Illinois at Urbana-Champaign.

Mathematics Education

- 01/2014 Math in the City: connecting the classroom to the local community through mathematical modeling, *Joint Mathematics Meetings, Baltimore, MD.*
2013 Co-organizer for the *Math in the City Workshop* for external faculty development.

Computational Algebra Software Development

Software developer for the computer algebra program *Macaulay2*. Software packages developed or contributed to:

- Generic Initial Ideal,
- Regularity,
- Symmetric Polynomials,
- Big Ideals,
- Normal Toric Varieties,
- SymbolicPowers,
- MacaulayPosets.